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 Studierichting: Informatica
 Oefeningenreeks 2B nr. 6.4

$$\text{Bewijs dat: } \text{Bgtan } \frac{1}{2} + \text{Bgtan } \frac{1}{5} + \text{Bgtan } \frac{1}{8} = \frac{\pi}{4}$$

$$\text{Stel } \Sigma = \text{Bgtan } \frac{1}{2}, \alpha = \text{Bgtan } \frac{1}{5}, \beta = \text{Bgtan } \frac{1}{8}$$

$$\Rightarrow \tan(\Sigma) = \frac{1}{2}, \tan(\alpha) = \frac{1}{5}, \tan(\beta) = \frac{1}{8}$$

$$\Rightarrow \Sigma + \alpha + \beta = \frac{\pi}{4}$$

$$\Rightarrow \tan(\Sigma + (\alpha + \beta)) = \tan\left(\frac{\pi}{4}\right) = 1$$

$$\text{Linkerzijde} = \frac{\tan \Sigma + \tan(\alpha + \beta)}{1 - \tan \Sigma \cdot \tan(\alpha + \beta)}$$

$$= \frac{\tan \Sigma + \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}}{1 - \tan \Sigma \cdot \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}}$$

$$= \frac{\frac{1}{2} + \frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \cdot \frac{1}{8}}}{1 - \frac{1}{2} \cdot \frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \cdot \frac{1}{8}}}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}}$$

$$= \frac{\frac{3}{6} + \frac{2}{6}}{\frac{6}{6} - \frac{1}{6}}$$

$$= \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

$$\text{Rechterzijde} = \tan \frac{\pi}{4} = 1$$

$$\Rightarrow \text{L.L} = \text{R.L}$$

Q.E.D.

References